TEMASEK POLYTECHNIC

SCHOOL OF INFORMATICS & IT

DIPLOMA IN CYBERSECURITY & DIGITAL FORENSICS

AY 2026/2027

PROJECT Report & Demonstration

(April Semester)

ENTERPRISE NETWORKING (CCF2C06)

SUBJECT LEVEL: 2

---

1. This project is an individual work.
2. This assessment is worth 40% of the subject total marks. The tasks for the given topology:
   a. Configure the given network in Packet Tracer based the specifications in Appendix A.
   b. For the given network topology, you are to research and propose ways to improve:
      - Performance
      - Reliability
      - Manageability
      - Security

      You are to put your research into a report **ONLY** and there is no need to implement them.
   c. You are required to demonstrate your work

3. The deliverables are as follows:
    a. Completed configuration in Packet Tracer for the given network topology
    a. A WORD report on the research and propose ways to improve the given network topology based on **ANY TWO** of the following 13 attributes:
        - Performance
            1. Bandwidth Optimization
            2. Latency Reduction
            3. Quality of Service (QoS)
            4. Network Monitoring
        - Reliability
            5. Uptime Guarantees
            6. Error Handling
            7. Failover Systems
        - Manageability
            8. Centralized Management
            9. Automation
            10. Policy Enforcement
        - Security
            11. Zero Trust
            12. Access Control
            13. Intrusion Detection / Prevention

4. The breakdown of marks (40%) award as follows:
    a. Report
        - Proposal: Research findings & Q&A (10%)
        - Contents and Layout (10%)
    b. Demonstration of assigned network topology configuration in Packet Tracer (10%) + Q&A (10%)

Deadlines

Demonstration - During workshop in Week 15
Report Submission: 31 July 5 PM (Friday)

Please take note of the following penalties with regards to late submissions:

- Late within 1 day: 10% deduction from absolute mark given for the assignment e.g 75 marks (100 marks max) becomes 65 marks (deduct 10% of 100 marks)
- Late > 1 day and up to 2 days: 20% deduction from absolute mark
- Late > 2 days: 0 marks awarded

Project Description

You are a newly hired network engineer of a networking consultancy company, Innova Corporation Ltd. You have just proposed the IP addressing scheme for the given topology (Toplogy_X or Topology_Y based on Project Part I). You are now task to complete the following for the assigned network topology:

1. Configure the Packet Tracer topology based on the specifications given in Appendix A for the respective network topology.
2. Your tutor will assign you to research and propose ways to improve the given network topology based on **TWO** of the following attributes. [**Tutors to note: For each class, there will be NOT more than 3 students assigned to each attribute below**]
   - Performance
     1. Bandwidth Optimization
     2. Latency Reduction
     3. Quality of Service (QoS)
     4. Network Monitoring
   - Reliability
     5. Uptime Guarantees
     6. Error Handling
     7. Failover Systems
   - Manageability
     8. Centralized Management
     9. Automation
     10. Policy Enforcement
   - Security
     11. Zero Trust
     12. Access Control
     13. Intrusion Detection / Prevention

---------- END OF DOCUMENT ---------

Appendix A.0 – Common Network Requirements

The following requirements are common for **ALL** topologies:

- All routers/switches hostname should be given meaning name to reflect either its role, location etc... or to differentiate from other routers/switches.
- All routers/switches to be secured and accessible **ONLY** via SSH on line vty 0 4. Using the domain **ENNK.COM.** Use 1024 bits or higher for crypto keys.
- Routing updates should be advertised ONLY when necessary to avoid security breaches.
- Use the enable secret password: **cisco123** for all routers/switches.
- All passwords should not be readable when **show run** command is executed on routers/switchers.
- All unused ports should be shut down for security reasons.
- Create a local user account (for remote access) with FULL configuration privileges in all routers & switches:
  - ✓ Username: admin
  - ✓ Password: cisco123
- There should **NOT** be any redundant configurations in routers/switches.
- For this project, we will **ignore** the configuration of Management VLAN 1 on switches.
- On all routers and switches, the following message should be displayed when user connects to it:

```
************************************************************************
* WARNING!                                                             *
*                                                                      *
* This is a private system. Unauthorized access is strictly prohibited.  *
* All activities on this device are logged and monitored. If you are   *
* not authorized to use this system, disconnect now.                   *
*                                                                      *
* Evidence of unauthorized use may be provided to law enforcement      *
* agencies.                                                            *
************************************************************************
```

Appendix A.1 – Topology_X Network Requirements

NOTE: Ensure you download the latest version of your topology

1. Use the IP address as proposed in Project Part 1.
2. Configure SMTP, HTTP and DNS at the Public Network.
   a. Create 3 email accounts: Andrew, Max and JohnLeong with password cisco123 at the Public Mail Server. Configure host machine PC0, PC8 and VLAN100 Wireless Laptop with one of each email account (Andrew, Max and JohnLeong) and test to show it is working.
   b. Create a default web page at the Public Web Server with the enlarge words "ISP Server" when user browse to it.
   c. Public DNS Server used by all host machines in the network. It should provide name translation for **all servers** in the network.
   d. Use only static routing at the ISP network.
3. All internal/private routers and switches time should be synchronized to the nearest NTP server.
4. All internal private router and switches should send logs at the **Informational** level (Level 6) or higher to the nearest syslog server.
5. Configure the internal private web server with url www.LeftServer.com with a default page showing the image:

6. All user host/end devices should acquire it's IP address from the nearest DHCP server (which is the one reached through traversing the least number of hops/router). If in doubt, check with your tutor.
7. Configure OSPF routing areas based on the labeled topology. The public network should not be able to gather information about the internal private networks. Minimum default route(s) should be used for the internal private network to exit to the public network. The exit **WAN 200.200.200.8/30** is a backup link to the public network. By default, all internal/private traffic should reach the public network via the **WAN 200.200.200.4/30** network.

8. Using the **Wireless LAN Controller0**, configure WIFI access for the VLAN networks connected to **Router22**.
9. Configure Dynamic PAT at **Edge Router** using the public pool:

   88.99.77.0/240

10. Internal/Private devices SHOULD be able to ping to all Internal and external devices.
11. External devices should NOT be able to ping to the internal devices.

---------- END OF **TOPOLOGY X** NETWORK REQUIREMENTS ---------

Appendix A.2 – Topology_Y Network Requirements

NOTE: Ensure you download the latest version of your topology

1. Use the IP address as proposed in Project Part 1.
2. Configure the following services:
   a. **DHCP - Net Admin / Reception** server to provide IP addresses to ALL end/host devices connected to **switch0** and **switch1**.
   b. **DHCP Server (Hostel Student/Warden)** server to provide IP addresses to ALL **VLAN800** and **VLAN900** end/host devices.
   c. **DHCP Server (Staff/Library)** server to provide IP addresses to ALL end/host devices connected to **switch11**, **switch12, switch 14 and switch 16.**
   d. **DHCP Server – Guest** server to provide IP addresses to ALL end/host devices connected to **switch 17.**
   e. **DHCP Server - Engineering Dept** server to provide IP addresses to ALL end/host devices connected to **VLAN 100, VLAN 200 and VLAN 300.**
   f. Create email accounts: **studentEmail, wardenEmail, libarianEmail, guestEmail and engineerEmail** with password **Cisco123** at the **SMTP server** located at the External network. Configure the email client with these accounts on any host machine in the respective network.
   g. The DNS server at the external network is used by all host machines in the network. It should provide name translation for all servers in the network topology.
3. Configure the external network Web server with url www.public.com with a default page showing the image:

4. Configure OSPF routing areas based on the labeled topology. The external network should not be able to gather information about the internal private networks. All internal network traffic should exit to the external network via WAN 100.100.100.20/30. The WAN network 100.100.100.24/30 is for backup and is used only when necessary.
5. Configure dynamic PAT at **Net-Admin** router to use the public pool: 111.111.111.0/29. Configure dynamic PAT at **Router7** router to use the public pool: 111.111.111.8/29.

6. Using the **Wireless LAN Controller0**, configure WIFI access for the Warden Wireless VLAN 800 network and the student Wireless VLAN 900.
7. Internal/Private devices SHOULD be able to ping to all Internal and external devices.
8. External devices should NOT be able to ping to the internal devices.

---------- END OF **TOPOLOGY Y** NETWORK REQUIREMENTS ---------

Rubrics

	Assessment Criteria	Grade A	Grade B	Grade C	Grade D/F
Report [20 Marks]	Proposal: Research Findings & Q&A	Exceptional research on TWO attributes. Provides actionable, technical improvements specifically for the assigned topology with a confident defense.	Solid research on two attributes. Proposals are technically sound but may be slightly generic. Able to answer the majority of Q&A questions.	Research is surface-level. Only one attribute is thoroughly explored, or the connection to the topology is weak. Struggles with technical Q&A.	Little to no research provided. Attributes are not from the approved list . Unable to justify the proposal.
	Report: Contents and Layout	Professional documentation. Includes full IP schemes, verification screenshots, and follows all Appendix A requirements. No formatting errors.	Well-organized report. Includes most IP data and configuration screenshots. Minor formatting inconsistencies or missing minor Appendix details.	Information is present but disorganized. Missing several key verification screenshots or IP addressing tables.	Incomplete report. Fails to document the required topology specifications, hostnames, or IP scheme.
Demonstration [20 Marks]	Demonstration: Configuration	Flawless implementation of all services (DHCP, DNS, SMTP, HTTP) and security (SSH, 1024-bit keys, WLC).	Most services functional. Minor errors in non-critical settings (e.g., banner typos or syslog level 6 synchronization errors).	Basic connectivity exists, but major services like WLC, Email, or Backup WAN links are non-functional.	Configuration is largely non-functional. Security requirements ignored (e.g., unused ports not shut down). Fails "The Cat Test".
	Demonstration: Technical Q&A	Displays mastery of the logic. Can explain OSPF area design, PAT pool logic, and troubleshooting steps fluently.	Shows good understanding. Can explain most configurations but requires prompting for complex routing or NAT/PAT logic.	Can explain basic setup (IPs/Hostnames) but fails to explain advanced logic like floating static routes or WLC mapping.	Cannot explain how the configuration works. Displays a significant lack of understanding regarding the network's operation.